

Numerical Audit Report

Synthetic Manuscript Audit Demo

This report flags numerical, statistical, and internal-consistency issues. It does not correct values — every item below requires manual verification and, where applicable, a decision by the author.

Status	Count
FAIL	5
FLAG	6
PASS	1

Arithmetic Reconciliation

3 flagged, 1 passed.

[FAIL] Components in group 'sv_type_counts' sum to 75, but stated total is 80

Location: sv_counts table

```
components: [('DEL', 40), ('DUP', 25), ('INV', 10)]
stated_total: 80
computed_total: 75
difference: -5
```

Suggested direction: Re-check whether all components are included, or whether the total was updated independently.

[FAIL] 'INV_confirmation_rate' reported as 90.0%, but $9/12 = 75.00\%$

Location: Table 3, INV row

```
numerator: 9
denominator: 12
reported_pct: 90.0
computed_pct: 75.0
difference_pp: -15.0
```

Suggested direction: Check whether numerator/denominator match what was actually used to compute the stated percentage.

[FAIL] 'DEL_confirmation_rate' reported as 92.5%, but $36/40 = 90.00\%$

Location: Table 3, DEL row

```
numerator: 36
denominator: 40
reported_pct: 92.5
computed_pct: 90.0
difference_pp: -2.5
```

Suggested direction: Check whether numerator/denominator match what was actually used to compute the stated percentage.

Statistical Validity

4 flagged, 0 passed.

[FLAG] 'INV_confirmation_rate' (9/12) is reported as a bare proportion with no confidence interval

Location: Table 3, INV row

```
numerator: 9
denominator: 12
```

Suggested direction: Consider reporting a Wilson score or Clopper-Pearson interval alongside the point estimate.

[FLAG] 'DUP_falsepos_confirmation_rate' (0/40) is reported as a bare proportion with no confidence interval

Location: Table 3, DUP false-positive row

```
numerator: 0
denominator: 40
```

Suggested direction: Consider reporting a Wilson score or Clopper-Pearson interval alongside the point estimate.

[FLAG] 'DUP_falsepos_confirmation_rate' is a degenerate sample (0/40) — any naive CI method (e.g. percentile bootstrap) will collapse to a single point and understate uncertainty

Location: Table 3, DUP false-positive row

```
numerator: 0
denominator: 40
reported_ci: None
reference_wilson_ci: (0.0, np.float64(0.0876))
reference_clopper_pearson_ci: (0.0, np.float64(0.0881))
reference_rule_of_three_upper_bound: 0.075
```

Suggested direction: Report Wilson score or Clopper-Pearson interval instead of a naive method for this degenerate case (values shown for comparison only — verify independently before use).

[FLAG] 'INV_confirmation_rate' is based on n=12 (< 30) and reported as a bare point estimate

Location: Table 3, INV row

```
numerator: 9
denominator: 12
```

Suggested direction: At this sample size the point estimate alone is likely to overstate precision; consider an exact interval and/or explicitly caveating n.

Cross-Section Consistency

1 flagged, 0 passed.

[FLAG] Metric 'total_strains' reported with 2 different values across 2 locations

Location: Abstract; Methods section 2.1

```
occurrences: [{'value': 12, 'location': 'Abstract', 'source': None}, {'value': 15, 'location': 'Methods section 2.1', 'source': None}]
```

Suggested direction: Confirm which value is current; if an update is intentional (e.g. dataset expanded), add an explicit note rather than leaving both values unreconciled.

Independence / Overlap

1 flagged, 0 passed.

[FAIL] 'calibration_set' and 'truth_set' are claimed independent but share 2 of 5/5 IDs

Location: Methods 2.1 / Methods 2.2

```
overlap_count: 2
overlap_sample: ['sv_003', 'sv_004']
set_a_size: 5
set_b_size: 5
```

Suggested direction: Re-derive the sets to be disjoint, or drop the independence claim and report the overlap.

Definition Drift

1 flagged, 0 passed.

[FLAG] Definition of 'CONFIRMED' at Supplementary Table 3 footnote differs from the canonical definition first seen at Methods section 2.3

Location: Supplementary Table 3 footnote

canonical_definition: A call is CONFIRMED if both breakpoint coordinates match ground truth within 50bp AND the SV type matches.

canonical_location: Methods section 2.3

this_definition: A call counts as CONFIRMED if the SV type matches ground truth, including PARTIAL coordinate matches.

similarity: 0.406

Suggested direction: Register a single canonical definition for 'CONFIRMED' and confirm every usage matches it.

Formula / Scoring Sanity

1 flagged, 0 passed.

[FAIL] Formula 'layer4_junction_score' is expected to vary across inputs but output std=0.007483 across 10 samples (range observed: 0.98-1.0)

Location: Layer 4 scoring module, all SVs

std: 0.007483314773547889

min: 0.98

max: 1.0

n: 10

spec_description: Breakpoint junction confidence score; should discriminate between well- and poorly-supported junctions across the dataset.

Suggested direction: Formula output is effectively constant — check for saturation (e.g. a component of the formula always evaluating to the same term).